

Module 5

Page Replacement Schemes

Page Replacement Policies

- A page replacement algorithm is needed to decide which page needs to be replaced when a new page comes in.
- These are used in operating systems that use paging for memory management.
- Page replacement is referred to a scenario in which a page from the main memory should be replaced by a page from secondary memory.
- Page replacement occurs due to page faults.

Demand Paging – Page Replacement Policies

- **The various page replacement algorithms like**

1. FIFO
2. Optimal page replacement
3. LRU
4. LRU-Approximation Page Replacement
5. Counting-Based Page Replacement

Demand Paging – Page Replacement Policies

- Page replacement is needed in the operating systems that use virtual memory using Demand Paging.
- In Demand paging, only a set of pages of a process is loaded into the memory.
- When a page that is residing in virtual memory is requested by a process for its execution, the Operating System needs to decide which page will be replaced by this requested page.
- This process is known as page replacement

Page Replacement Policies - FIFO

- Maintain a queue of all the pages that are in the memory currently.
 - The oldest page in the memory is at the front-end of the queue and
 - The most recent page is at the back or rear-end of the queue.
- Whenever a page fault occurs, the operating system looks at the front-end of the queue to know the page to be replaced by the newly requested page.
- It also adds this newly requested page at the rear-end and removes the oldest page from the front-end of the queue.

Page Replacement Policies - FIFO - Example

- Consider the page reference string as 3, 1, 2, 1, 6, 5, 1, 3 with 3-page frames. Let's try to find the number of page faults.

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

← Page Frames

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

- Consider the page reference string as 3, 1, 2, 1, 6, 5, 1, 3 with 3-page frames. Let's try to find the number of page faults.

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

← Page Frames

← Faults Occurred

← Replaced Page

Page Replacement Policies – FIFO – Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

3

← Page Frames

Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

	1
3	3

← Page Frames

Y	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2
	1	1
3	3	3

← Page Frames

Y	Y	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2
	1	1	1
3	3	3	3

← Page Frames

Y	Y	Y	Hit

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2	2
	1	1	1	1
3	3	3	3	6

← Page Frames

Y	Y	Y	Hit	Y
				3

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2	2	2
	1	1	1	1	5
3	3	3	3	6	6

← Page Frames

Y	Y	Y	Hit	Y	Y
				3	1

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2	2	2	1
	1	1	1	1	5	5
3	3	3	3	6	6	6

← Page Frames

Y	Y	Y	Hit	Y	Y	Y
				3	1	2

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3	← Page Referenced
		2	2	2	2	1	1	← Page Frames
	1	1	1	1	5	5	5	
3	3	3	3	6	6	6	3	
Y	Y	Y	Hit	Y	Y	Y	Y	← Faults Occurred
				3	1	2	6	← oldest/Replaced Page

Page Replacement Policies – FIFO - Example

- Number of Hits: 1 (Count only Hit)
- Page Faults: 7 (Count only Y)
- Hit Rate: 12.5% ($\text{No. of Hits} / \text{No. of References} = 1 / 8$)

Page Replacement Policies - FIFO

Belady's Anomaly

- Less number of frames, then the number of page faults is more.
- The **number of frames is inversely proportional to the number of page faults**.
- But some scenarios lead to - a situation occurs when this concept is reversed.
- i.e., The **number of frames becomes directly proportional to page faults**
- **increase the number of frames, page faults also increase.**
- This direct proportionality of frames and page number is known as "Belady's Anomaly".

Page Replacement Policies - FIFO - Example2

- In Previous example let's increase the Frames by 1. so total frames will be 4 and having the same referenced pages.
- Pages Referred - 3, 1, 2, 1, 6, 5, 1, 3
- No. of Frames - 4

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

3

← Page Frames

Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

	1
3	3

← Page Frames

Y	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2
	1	1
3	3	3

← Page Frames

Y	Y	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2
	1	1	1
3	3	3	3

← Page Frames

Y	Y	Y	H

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

				6
		2	2	2
	1	1	1	1
3	3	3	3	3

← Page Frames

Y	Y	Y	H	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

				6	6
		2	2	2	2
	1	1	1	1	1
3	3	3	3	3	5

← Page Frames

Y	Y	Y	H	Y	Y
					3

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

				6	6	6
		2	2	2	2	2
	1	1	1	1	1	1
3	3	3	3	3	5	5

← Page Frames

Y	Y	Y	H	Y	Y	H
					3	3

← Faults Occurred

← Replaced Page

Page Replacement Policies - FIFO - Example

3	1	2	1	6	5	1	3	← Page Referenced
				6	6	6	6	← Page Frames
		2	2	2	2	2	2	
	1	1	1	1	1	1	3	
3	3	3	3	3	5	5	5	
Y	Y	Y	H	Y	Y	H	Y	← Faults Occurred
					3		1	← Replaced Page

Page Replacement Policies - FIFO - Example2

- Pages Referred - 3, 1, 2, 1, 6, 5, 1, 3 = 8
- No. of Frames – **4**
- No. of Hits – **2**
- No. of Faults – **6**
- Hit Ratio = $2/8 = \mathbf{25\%}$

Page Replacement Policies - FIFO - Example 3

- Consider the reference string = 3, 2, 1, 3, 4, 1, 6, 2, 4, 3, 4, 2, 1, 4, 5, 2, 1, 3, 4
 - Number of Frames = 3
 - Find Page faults and Hit Ratio

Page Replacement Policies – Least Recently Used (LRU)

- Keeps the track of usage of pages over a period.
- Works on the basis of the principle of locality of a reference which states that a program has a tendency to access the same set of memory locations repetitively over a short period of time.
- Pages that have been used heavily in the past are most likely to be used heavily in the future also.
- when a page fault occurs, then the **page that has not been used for the longest duration of time** is replaced by the newly requested page

Page Replacement Policies - LRU - Example 1

- Consider the page reference string as 3, 1, 2, 1, 6, 5, 1, 3 with 3-page frames. Let's try to find the number of page faults.

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

← Page Frames

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

3

← Page Frames

Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

	1
3	3

← Page Frames

Y	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2
	1	1
3	3	3

← Page Frames

Y	Y	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2
	1	1	1
3	3	3	3

← Page Frames

Y	Y	Y	H

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2	2
	1	1	1	1
3	3	3	3	6

← Page Frames

Y	Y	Y	H	Y
				3

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example 1

3	1	2	1	6	5	1	3
---	---	---	---	---	---	---	---

← Page Referenced

		2	2	2	5
	1	1	1	1	1
3	3	3	3	6	6

← Page Frames

Y	Y	Y	H	Y	Y
				3	2

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example 1

3	1	2	1	6	5	1	3
---	---	---	---	---	---	---	---

← Page Referenced

		2	2	2	5	5
	1	1	1	1	1	1
3	3	3	3	6	6	6

← Page Frames

Y	Y	Y	H	Y	Y	H
				3	2	

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example 1

3	1	2	1	6	5	1	3
---	---	---	---	---	---	---	---

← Page Referenced

		2	2	2	5	5	5
	1	1	1	1	1	1	1
3	3	3	3	6	6	6	3

← Page Frames

Y	Y	Y	H	Y	Y	H	Y
				3	2		6

← Faults Occurred

← Replaced Page

Page Replacement Policies - LRU - Example 1

- Pages Referred - 3, 1, 2, 1, 6, 5, 1, 3 = 8
- No. of Frames – 3
- No. of Hits – 2
- No. of Faults – 6
- Hit Ratio = $2/8 = 25\%$

Page Replacement Policies – Optimal Page Replacement

- Optimal page replacement is the best page replacement algorithm as this algorithm results in the least number of page faults.
- The pages are replaced with the ones that will not be used for the longest duration of time in the future.
- The pages that will be referred farthest in the future are replaced in this algorithm.

Page Replacement Policies - Optimal - Example 1

- Consider the page reference string as 3, 1, 2, 1, 6, 5, 1, 3 with 3-page frames. Let's try to find the number of page faults.

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

← Page Frames

← Faults Occurred

← Replaced Page

Page Replacement Policies – Optimal – Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

3

← Page Frames

Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - Optimal - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

	1
3	3

← Page Frames

Y	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - Optimal - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2
	1	1
3	3	3

← Page Frames

Y	Y	Y

← Faults Occurred

← Replaced Page

Page Replacement Policies - Optimal - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2
	1	1	1
3	3	3	3

← Page Frames

Y	Y	Y	H

← Faults Occurred

← Replaced Page

Page Replacement Policies - Optimal - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

Which Page is Far?



		2	2	6
	1	1	1	1
3	3	3	3	3

← Page Frames

Y	Y	Y	H	Y
				2

← Faults Occurred

← Replaced Page

Page Replacement Policies - Optimal - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

Which Page is Far?



		2	2	6	5
	1	1	1	1	1
3	3	3	3	3	3

← Page Frames

Y	Y	Y	H	Y	Y
				2	2

← Faults Occurred

← Replaced Page

Page Replacement Policies - Optimal - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2	6	5	5
	1	1	1	1	1	1
3	3	3	3	3	3	3

← Page Frames

Y	Y	Y	H	Y	Y	H
				2	2	

← Faults Occurred

← Replaced Page

Page Replacement Policies - Optimal - Example 1

3	1	2	1	6	5	1	3
----------	----------	----------	----------	----------	----------	----------	----------

← Page Referenced

		2	2	6	5	5	5
	1	1	1	1	1	1	1
3	3	3	3	3	3	3	3

← Page Frames

Y	Y	Y	H	Y	Y	H	H
				2	2		

← Faults Occurred

← Replaced Page

Page Replacement Policies - Optimal - Example 1

- Pages Referred - 3, 1, 2, 1, 6, 5, 1, 3 = 8
- No. of Frames – 3
- No. of Hits – 3
- No. of Faults – 5
- Hit Ratio = $3/8 = 37.5\%$

Page Replacement Policies - Optimal - Example 2

- Consider the Reference String – 7 0 1 2 0 3 0 4 2 3 0 3
2 1 2 0 1 7 0 1
 - No. of Frames - 4
 - Find the number of page faults and hit ratio.

Page Replacement Policies - Optimal - Example 1

- No. of Faults – 8
- Hit Ratio = $11/20$

LRU-Approximation Page Replacement

- Some systems provide no hardware support
- Many systems provide support in the form of a **reference bit**
- Reference bits are associated with each entry in the page table.
- Initially, all bits are cleared (to 0) by the operating system. As a process executes, the bit associated with each page referenced is set (to 1) by the hardware.

Additional-Reference-Bits Algorithm

- Recording the reference bits at regular intervals
- An 8-bit byte for each page in a table in memory is used
- At regular intervals (say, every 100 milliseconds), a timer interrupt transfers control to the operating system.

Additional-Reference-Bits Algorithm

- The 8-bit shift registers contain the history of page use for the last eight time periods.
- If the shift register contains 00000000, for example, then the page has not been used for eight time periods.
- In the extreme case, the number can be reduced to zero, leaving only the reference bit itself. This algorithm is called the **second chance page-replacement algorithm**.

Additional-Reference-Bits Algorithm

- In the extreme case, the number can be reduced to zero, leaving only the reference bit itself. This algorithm is called the **second chance page-replacement algorithm**.

Second chance page-replacement algorithm.

- It is also referred as **clock** algorithm
- One way to implement the second-chance algorithm is as a circular queue.
- A pointer indicates which page is to be replaced next.
When a frame is needed, the pointer advances until it finds a page with a 0 reference bit.

Second chance page-replacement algorithm.

- Once a victim page is found, the page is replaced, and the new page is inserted in the circular queue in that position.
- Notice that, in the worst case, when all bits are set, the pointer cycles through the whole queue, giving each page a second chance.

Enhanced Second chance algorithm

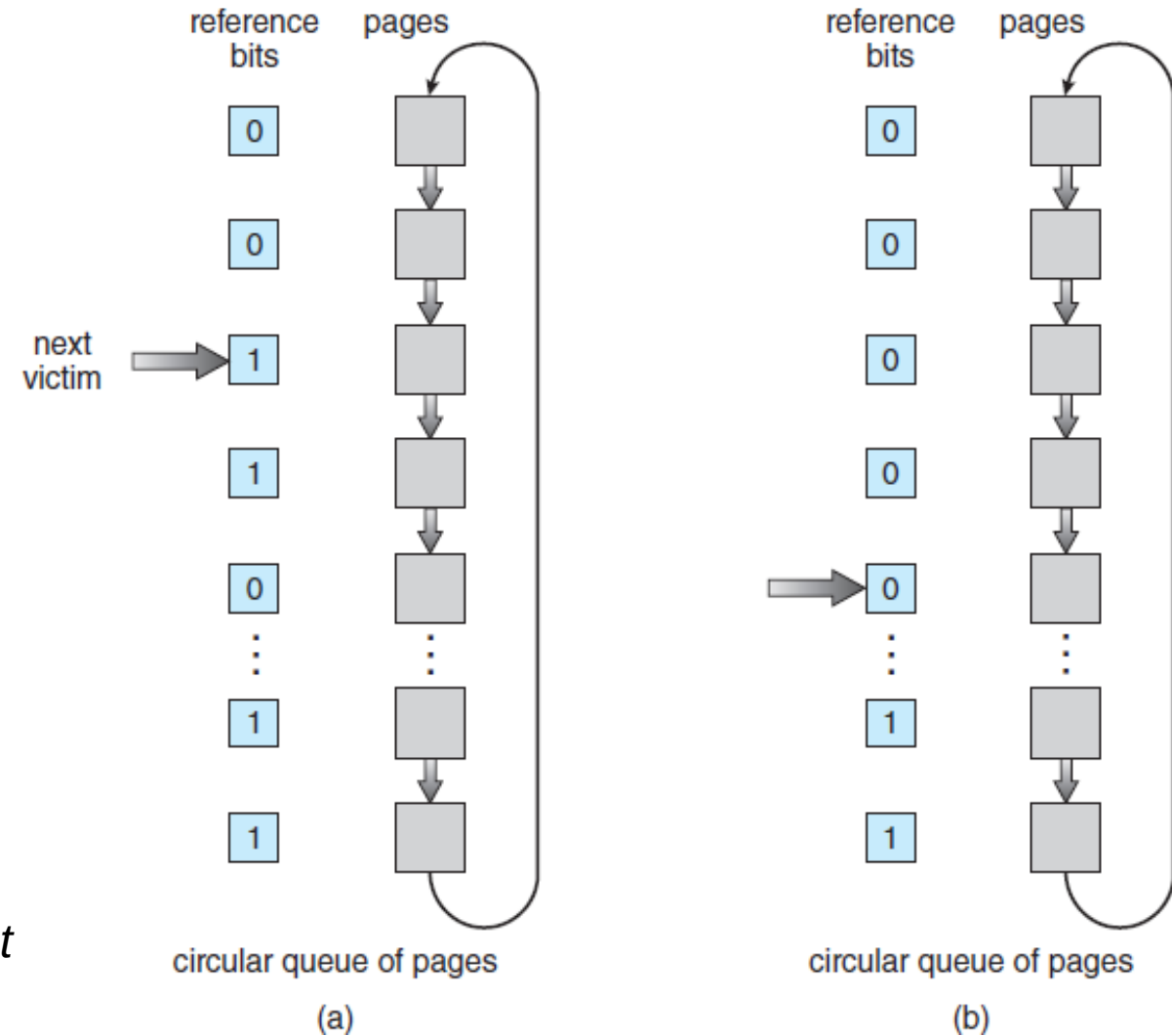


Figure Second-chance (clock) page-replacement algorithm

Second chance page-replacement algorithm- Ex.01

Consider the given string: 2 3 2 1 5 2 4 5 3 2 5 2

Frame size: 03

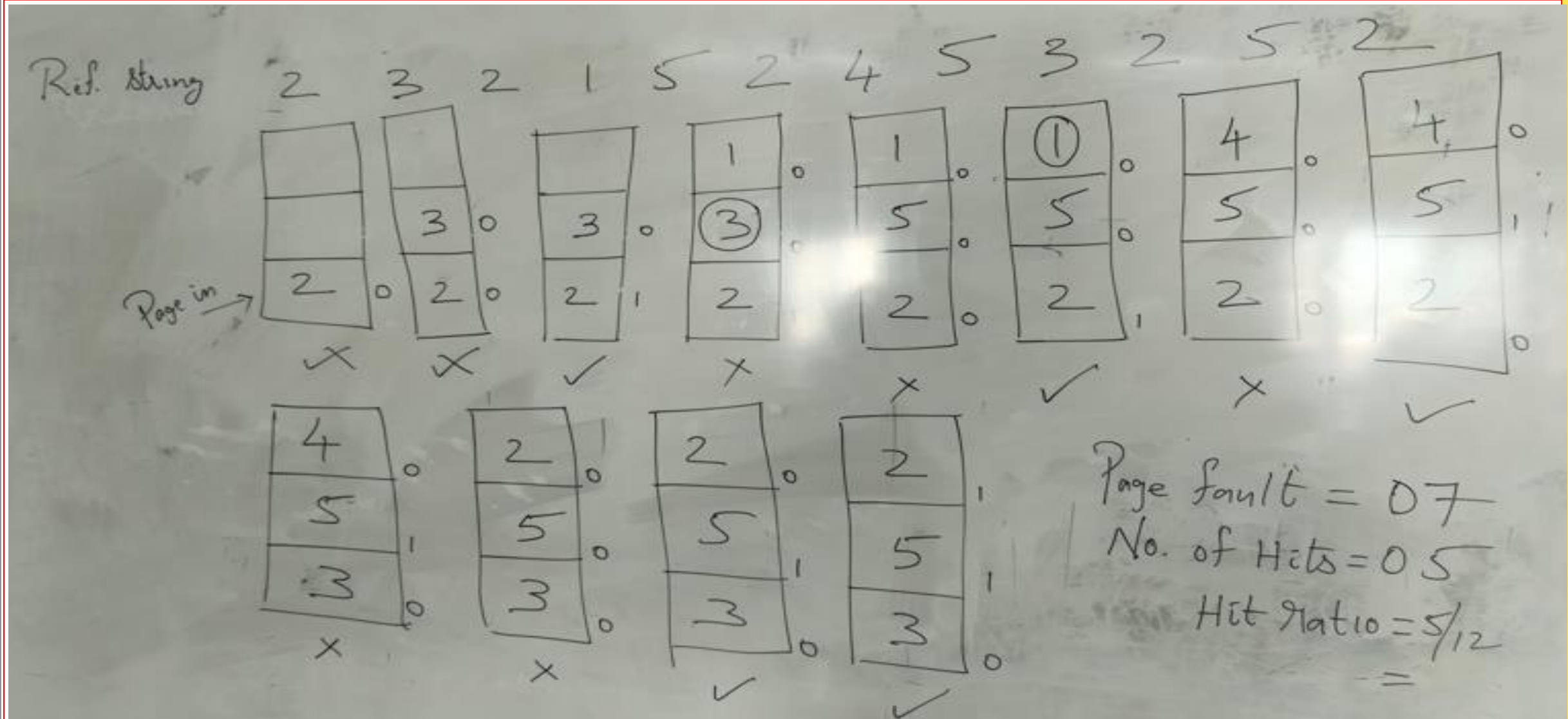
Find the page fault and hit ratio?

Answer

Page fault = 7

Hit ratio = $5/12$

Second chance page-replacement algorithm- Ex.01



Second chance page-replacement algorithm- Ex.02

Consider the given string:

4 5 5 5 6 7 8 6 7 8 4 1 5 2 4 4 1

Frame size: 04

Find the page fault and hit ratio?.

Enhanced Second chance algorithm

- Enhance the second-chance algorithm by considering the **reference bit** and the **modify bit** as an ordered pair. With these two bits, we have the following four possible classes:
 1. (0, 0) neither recently used nor modified—best page to replace
 2. (0, 1) not recently used but modified—not quite as good, because the page will need to be written out before replacement
 3. (1, 0) recently used but clean—probably will be used again soon
 4. (1, 1) recently used and modified—probably will be used again soon, and the page will need to be written out to secondary storage before it can be replaced

Enhanced Second chance algorithm

- Each page is in one of these four classes. When page replacement is called for, we use the same scheme as in the clock algorithm; but instead of examining whether the page to which we are pointing has the reference bit set to 1, we examine the class to which that page belongs.

Counting-Based Page Replacement

- a counter is used to count the number of references that have been made to each page
- The **least frequently used (LFU)** page-replacement algorithm requires that the page with the smallest count be replaced.
- The **most frequently used (MFU)** page-replacement algorithm is based on the argument that the page with the smallest count was probably just brought in and has yet to be used.